TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Robotic Bin-Picking			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Robot Vision and Production Automation Technologies	Robotics/Automation Machinery	Robot Control and Intelligence Technologies
Concept of Proposed Technology	Core concept: Robotic Bin-Picking: - Singulation of chaotically stored objects by using a robot, followed by precise placement of the picked object at a target location. - The Bin-Picking software receives a 3D point cloud from a 3D sensor (typically mounted above the bin) and outputs a collision-free robot path for grasping (pick object from a cluttered bin) and placement. - See videos (2) and (3) Submodules: - 6D Object Pose Estimation - Grasps Generation - Path Planning			

	- Collision Avoidance - Physics Simulation - Automatic Optimization for autonomy and performance improvements Limitations of current Bin-Picking solutions: Despite numerous good solutions on the market, Bin-Picking is not yet in widespread use. The reasons for this are manifold: - Time-consuming manual teach-in by experts (no plug-and-produce) - Lack of robustness - No complete emptying of the bin - Challenging objects (reflective, flat, etc.) - Lack of accuracy - Too high cycle time - … Central goal: Improvement of autonomy and performance of robotic Bin-Picking applications Main functions: - Virtual Feasibility Studies: Conducted for Bin-Picking to enhance transparency and reduce risks for end users at an early stage in the project. - Training Data Generation: Generation of synthetic data for training of AI-based deep neural network models specific to Bin-Picking applications. - Optimization: Focused on improving cell layout and gripper finger design to enhance the overall performance of the Bin-Picking cell. Potential uses of the proposed technology: - Automatic singulation of chaotically stored objects with a robot instead of manual handling - Reduced configuration effort for new objects for labor (automatic configuration by means of simulations and machine learning) - Utilization of the physics simulation and optimizations routines for performance improvements of the application - Increase transparency at an early project stage and reduce risks via virtual performance evaluation of the application
Rationale for Joint R&D Project	Trends and needs for R&D project (1): - The market for Bin-Picking solutions is expanding rapidly. However, existing systems often require manual optimizations for each part on-site by an expert. By integrating self-configuring, AI-based solutions, these systems can be deployed by non-professionals, enabling easy scalability and enhancing economic efficiency. - Robot cell layouts and gripper fingers are typically designed by experts. This limits the scalability to new applications. Suboptimal designs can lead to decreased performances

	regarding success rates, cycle times, etc. - Current Bin-Picking solutions often do not leverage knowledge or experience from simulated grasping trials. Utilizing this information can be highly advantageous for prioritizing easily graspable objects, avoiding entanglements, and ensuring that objects do not displace relative to the gripper during grasping. This is crucial for achieving precise object placements, a typical requirement in industrial processes. - In the research community, there is a strong trend towards model-free, learning-based (AI) solutions. While these solutions excel at generalizing to new objects, they often lack the precision needed for object placement—a typical requirement in industrial processes. Model-based solutions, on the other hand, can achieve precise object placements. Additionally, physics simulations can enhance model-based Bin-Picking solutions through synthetic data generation for deep neural networks, optimizations, and virtual feasibility studies. Collaboration: - Partners should provide real use-case form practical scenarios from industry - Provide relevant testing equipment (bin, objects, …) - Give relevant performance parameters (e.g., cycle time < 10 s, very high bin clearance rate, object placement variance < 2 mm, …) - Partner should give feedback on required improvements of the system during the project - Partner should give support regarding design choices of the system during the project
Key Words	1. Bin-Picking 2. Robotic Grasping and Manipulation 3. Simulations 4. Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) 5. Optimizations

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level 3-4	**End of the Project**	Level 6-7
R&D Objectives	Overall Goal: Increase autonomy and performance of robotic Bin-Picking applications by means of simulations, machine learning, and optimizations Modules: - Development and improvement of physics simulations for robotic Bin-Picking applications via modeling the robot movement, 3D sensor, and grasping principles (suction grippers, clamping grippers, …) - Development of a framework for virtual feasibility studies for robotic Bin-Picking for automatic performance evaluations for new objects - Synthetic training data generation for the training of deep neural networks for 6D object pose estimation and grasp pose selection - Automatic configuration of the software for new objects without any human intervention by means of simulations and machine learning - Automatic optimization of the gripper finger and cell design for performance improvements			
R&D Target Specifications	Virtual Feasibility Studies: Develop a framework to get output performance metrics for specific Bin-Picking use-cases and objects: - Grasp success rate - Placement success rate - Placement accuracy - Bin clearance rate - Number of entanglements - Cycle times (picks per hour) - … Automatic configuration for new objects without any human intervention: - Fully automated pipeline for synthetic data generation, deep neural network training, and testing (performance evaluation) - Reduction of the manual configurations effort (parameterization and testing time 3 h → 3 min) - Easy usability by layperson (no expert knowledge needed) Optimization of the gripper design and cell layout: - Reduction of the cycle time of the Bin-Picking application - Performance improvements due to superior gripper finger design (higher grasp success rates, more precise object placements, …)			

R&D Plan	Phase I (Pre-study): months 1 - 6 - Objective: Technical pre-study to confirm feasibility; prepare for Phase II - Use-case refinement together with all partners - Collection of all relevant boundary conditions - Selection and provision of representative objects for conducting the Bin-Picking tests - Documentation of requirements and success criteria - Demonstrate proof-of-concept: Simple optimization example - Demonstrate proof-of-concept: Set up physics simulation framework and demonstrate proper robot movements and gripping behavior (suction gripper, clamping gripper, …) - Demonstrate proof-of-concept: Present 6D object pose estimation results for example objects provided from the Korean SME partner Phase II (Main R&D): months 7 - 36 Work packages: WP 1: Development of realistic Bin-Picking simulations (months 7 - 18) - Set up digital twin of the Bin-Picking application (bin, objects, robot, 3D sensor, robot cell, obstacles, …) - Realistic grasping with different gripper types (suction cup gripper, parallel-jaw gripper, …) - Path planning: Implement solution for proper robot movements for grasping and placement of the objects - Implementation of data logging routines WP 2: Development and testing of routines for automatic grasps generation (months 12 - 18) - For suction cup grippers with multiple suction cups - For clamping grippers with multiple fingers WP 3: Development and testing of routines for automatic object configuration (months 18 - 32) - Concept specification for automatic object configuration (6D object pose and grasp pose) - Testing in simulation - Performance evaluation in simulation - Evaluation of configuration efforts (manual vs. automatic pipeline) WP 4: Development of optimization routines (months 18 - 32) - Development of optimization routines for automatic gripper finger optimization - Development of optimization routines for automatic cell design optimization WP 5: Improvement of the Sim-To-Real Transfer (months 18 - 32) - Evaluation of the influence of different randomization parameters on the real-world

	performance - Evaluation of the performance of different Suim-to-Real techniques on the performance of the deep neural networks for pose estimation and grasp pose selection (Domain Randomization vs. Domain Adaptation) WP 6: Real-world demonstrator setup (months 18 - 36) - Evaluation of the algorithms in the real-world demonstrator - Fine-tuning of software and hardware for performance improvements - Final performance evaluation - Video documentation Milestones: - MS 1: All algorithms developed, ready for deployment to the real-world demonstrator (month 24) - MS 2: Real-world demonstrator running, algorithms optimized, and all performance metrics identified for all relevant objects (month 36)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Core role: Software development Resources: - Dedicated project team - Lab facilities with robots, grippers, and 3D sensors - Simulation software and infrastructure
Expected R&R of Korean SME Partner	- Provide use-cases (Bin-Picking applications) - Provide specific objects for testing - Give requirements regarding performance - Build real-world demonstrator (robot, grippers, bins, objects, 3D sensors, …) to test the software solutions developed - Documentation of the results and the output performance metrics

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes: Technological achievements: - Simulation Benefits: Simulations promote the broader adoption of robotic and AI-based handling solutions. - Automatic configuration of new objects without any human intervention (no expert knowledge needed and easy usability by non-professionals) - Automated Routines: Simulations facilitate automatic optimization of cell layouts and gripper finger designs. - Automatic Performance Evaluations: Virtual feasibility studies, based on a digital twin of the robotic Bin-Picking cell, enable automatic assessments of key performance metrics such as grasping success rate, accuracy, bin clearance rate, and cycle time for new objects. Commercialization potential: - Reusable software modules (gripper optimization, cell optimization, synthetic data generation and deep neural network training, …) for integration by robot OEMs and system integrators - Clear business case: Addresses labor shortages, ergonomics, reliability, quality, … Broader industrial or economic impact: - Easy deployment of the self-configuring algorithms by non-professionals - No experts are needed for configuring new objects (enables easy scalability and enhances economic efficiency) - No experts are needed for designing new gripper fingers and cell layouts (enables easy scalability and enhances economic efficiency)

6. References

(1) Kilian Kleeberger, Richard Bormann, Werner Kraus, and Marco F. Huber, "A Survey on Learning-Based Robotic Grasping," Current Robotics Reports, vol. 1, no. 4, pp. 239–249, 2020.

(2) Video YouTube trade fair (Control 2022 in Stuttgart): https://youtu.be/3hq0L9bc_T0?si=1XTcl4v05ocHKUO_

(3) Video YouTube virtual feasibility studies: https://youtu.be/PP3Dqe8dCUo?si=LczWNR4w6oUj9VVw

☐ Terms:

1) Industrial Technology Classification

Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization